Assignment 3: ETL with Bash

Learning Objectives:

In this exercise, you will learn about ETL implementation using shell scripting language to perform the following tasks:

1. Ingest and extract data from remote sources
2. Transform data using shell scripting
3. Load data into an on-premise relational database (Postgres) using a Python script
4. Schedule to run the job at a preset time using crontab

Due Date: February 9, 2025

Evaluation: 10% of the total grade

Instructions

You are provided with a link to a remote dataset for this assignment. You are asked to create a Bash script to perform the Extract-Transform-Load process 0n the given dataset through following tasks:

1. Create a Bash script file (hint: touch filename.sh)
2. Open the file in your favorite text editor (nano is the easiest)
3. Initial the bash script (shebang) and define some variables
4. Start the ETL process by downloading COVID-19 data from the following URL (hint: curl / wget): (https://raw.githubusercontent.com/owid/covid-19-data/master/public/data/owid-covid-data.csv)
5. Transform data by selecting specific columns (1, 2, 4, and 5) (hint: cut -d -f)
6. Load the transformed data into a RDBMS, such as Postgres, by calling a python script which creates a database and a table (COVID) with column names matching the source before the data load. (hint: COPY table_name FROM /path/to/file.csv)
7. Run the Bash script to perform the ETL. Log/print appropriate messages throughout the process.
8. Create a crontab to run the ETL job everyday at 10 minutes past midnight.

Note: Submit your code and report with embedded screenshots to substantiate your work.

Scoring Rubric

Category	Criteria	Scores
Bash script	Well organized and easy to follow functional shell script with appropriate comments, includes shebang, appropriate permissions	4
Python code	Well organized and clean code wit appropriate comments, correct implementation of schema, successful data load	3
Cron table	Appropriately set cronjob schedule	1
Report	A well-organized report containing an executive summary followed by properly labeled sections showing results, screenshots and brief explanation of observations	2